

Praveensankar Manimaran

PhD Research Fellow

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Bio

Praveen is about to complete his PhD at the University of Oslo, Norway, where he has worked on privacy problems in verifiable credentials systems. Praveen has expertise in developing new research directions, designing, developing, and implementing solutions for complex research problems.

Education

Ph.D. - Informatics, *University of Oslo* 2020 Oct – 2026 Sep

- **Research in Privacy** - Developed privacy-preserving protocols for verifiable credentials systems using zero-knowledge proofs and blockchain.
- **Teaching Experience**: IN5020 - Distributed Systems (2022, 2023), IN9420 - Distributed Blockchain Technologies (2022, 2023).
- Installed and maintained Norway's EBSI pilot node (2022-2023).
- **Advisors**: Roman Vitenberg, Leander Jehl.

M.Tech. - Computer Science and Engineering 2018 Jul – 2020 Jun

National Institute of Technology Puducherry

- CGPA: 9.82, **Gold Medalist**

B.E. - Computer Science and Engineering, *PSG College of Technology* 2012 Jul – 2016 May

Professional Experience

Research and Engineering Scientist, *Zoho Corporation* 2025 Dec – 2026 Jul

- Designed systems using Large Language Models and ML technologies to solve business-critical problems.

Software Engineer, *Accolite Software India Pvt Ltd* 2016 Jul – 2017 Mar

- Worked on web development and used technologies associated with the .NET framework.

Academic Services

PETS 2027 - Reviewer

Publications

zkRevoke: Configurable Untraceability for Verifiable Credentials using ZKPs,

Proceedings on Privacy Enhancing Technologies (2026)

- Formalizes the privacy issuer of traceability of revocation status in the verifiable credentials system.

- Proposes *zkRevoke*, a revocation protocol that uses a custom-built Zero-Knowledge Proof circuit to address the privacy issue.
- *zkRevoke* outperforms existing protocols in terms of required bandwidth for issuers and holders.
- Link: <https://petsymposium.org/popets/2026/popets-2026-0086.php> 

Decentralization Trends in Identity Management: From Federated to Self-Sovereign Identity Management Systems, *Computer Science Review*, Volume 58, 2025

- Reviews Federated and Self-Sovereign Identity systems from a conceptual point of view.
- Proposes a generic framework for Identity systems and analyzes the architectures and workings of (a) IOTA Id, (b) Indy, and (c) eIDAS v1.0.
- Highlights differences in decentralization and privacy strategies across these systems.

Prevoke: Privacy-Preserving Configurable Method for Revoking Verifiable Credentials, *2024 IEEE International Conference on Blockchain (Blockchain)*

- *Prevoke* presents key privacy challenges in the revocation protocol and proposes a novel solution that utilizes Merkle Tree Accumulators, BBS Signatures, Bloom Filters and Smart Contract.
- *Prevoke* also proposes a two-phase verification protocol that optimizes the performance of verification. Most of the valid credentials go through fast verification. Only revoked and a handful of valid credentials would go through slow verification.

Research Adoption

EBIP: Secure Privacy-Preserving Revocation of Verifiable Credentials, [<https://hub.ebsi.eu/ebips>]

- Discloses a security vulnerability in EBSI's Dynamic Status List, a revocation protocol for verifiable credentials that addresses the privacy issue of holder's traceability.
- Adopts *zkRevoke*, a ZKP-based revocation protocol developed during my PhD.

Projects

zkRevoke, *Implementation*

2024 Aug – 2025 May

- Implemented *zkRevoke* and integrated *zkRevoke* into an in-built VC ecosystem.
- Built a custom ZKP circuit based on Groth16 ZKP scheme using the gnark library.
- Benchmarked the performance of *zkRevoke* and an existing protocol to perform the comparison.
- **Languages:** Golang, Solidity
- **Github:** <https://github.com/praveensankar/zkRevoke> 

Prevoke , *Implementation*

2023 Aug – 2024 Jul

- Implemented *Prevoke* and integrated *Prevoke* into an in-built VC ecosystem.
- **Workflows:** 1) issuance, 2) revocation, 3) VP construction and sharing, and 4) VP verification.
- In addition, deployed a Smart Contract on a Private Blockchain using Ganache, hosted in NREC.
- The entities are hosted as servers and geographically distributed via NREC.
- **Results:** Analyzed the performance, latency, and cost of *Prevoke* for different workloads.
- **Languages:** Golang, Solidity, Python
- **Github:** <https://github.com/praveensankar/Prevoke>  .